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Bihar Engineering University, Patna

B.Tech 2nd Semester Exam., 2014

Subject: ELEMENTS OF MECHANICAL ENGINEERING

Code: 021201

Time: 3 hours

Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Answer the following/Choose the correct option (any seven): **[2×7=14]**

(a) Give two examples of primary sources of energy.

(b) Zeroth law of thermodynamics forms the basis of _____ measurement.

(i) pressure
(ii) temperature
(iii) volume
(iv) work

(c) Difference between the pressure of a fluid and the pressure of atmosphere is called as

(i) Absolute pressure
(ii) Barometric pressure
(iii) Gauge pressure
(iv) Vacuum pressure

(d) Heat supplied to a system equals the work done in case of a non-flow process carried out

(i) isochorically
(ii) isobarically
(iii) adiabatically
(iv) Isothermally

(e) Which of the following fittings is a boiler mounting?

(i) Superheater
(ii) Feed check valve
(iii) Economiser
(iv) Air preheater

(f) The number of valves in a single-cylinder two-stroke petrol engine is

(i) one
(ii) two
(iii) four
(iv) zero

(g) How many degrees a crankshaft rotates in completing the cycle of two-stroke engine?

(h) The condenser in which there is direct contact between the steam and the cooling fluid should be

(i) surface condenser
(ii) jet condenser
(iii) evaporative condenser
(iv) All of the above

(i) 1 ton of refrigeration effect is equal to

(i) 3.5 kJ/s
(ii) 55 kJ/s
(iii) 200 kJ/s
(iv) 210 kJ/s

(j) The ability of a material to resist wear is called

(i) strength
(ii) toughness
(iii) hardness
(iv) brittleness

Q.2 Solve both questions : **[14]**

(a) Explain the merits and demerits of renewable and non-renewable sources of energy.

(b) What is the origin of biomass energy? Give the main advantages and disadvantages of biomass energy.

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Q.3 Solve both questions : **[14]**

(a) Differentiate between the following:
(i) Intensive and Extensive property
(ii) Point and Path function

(b) A mass of gas is compressed in a quasi-static process from 80 kPa, 0.1 m^3 to 0.4 MPa, 0.03 m^3. Assuming that the pressure and volume are related by pv^n= constant, find the work done by the gas system.

Q.4 Solve both questions : **[14]**

(a) Classify the different types of boilers. Differentiate between fire-tube and water-tube boilers.

(b) Classify mountings into safety fittings and control fittings. Explain with neat sketch any one of each.

Q.5 Solve both questions : **[14]**

(a) Derive an expression for air standard efficiency of Otto cycle.

(b) Differentiate between two-stroke and four-stroke engines.

Q.6 Solve both questions : **[14]**

(a) With neat sketch, explain briefly the working principle of an impulse turbine.

(b) Write the application of gas turbine. What are the advantages and disadvantages of gas turbine over IC engine?

Q.7 Solve both questions : **[14]**

(a) Why is steam condenser used in thermal power plant?

(b) Describe with neat sketch the construction and working of a nuclear power plant.

Q.8 Solve both questions : **[14]**

(a) Explain Bell-Coleman air refrigeration cycle.

(b) Explain with neat sketch, the working of vapour compression refrigeration system.

Q.9 Solve both questions : **[14]**

(a) What is plain carbon steel? Give classification of plain carbon steels with their important properties.

(b) What are the objectives of heat treatment? Explain the heat treatment process by tempering.

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